YEAR 7 · SCIENCE · MODELS · CHALLENGE HOMEWORK

# Why Is Mr Bowen Still Awake?

Half-life with coffee, an interview at home, carbon dating and the Sun



**Name:** _____ **Class:** _____

**Date given:** _____ **Date due:** _____

## How to do this homework

- **Part A:** Mr Bowen's coffee. It shows you the method.
- **Part B:** interview an adult at home. **Do it at the weekend.**
- **Part C:** carbon dating. **Part D:** the Sun. **Part E:** caffeine and sleep, and a letter.
- You need a **calculator**, a **ruler**, and a **phone or computer**.
- Answer in **full sentences**. "50 mg" is not a sentence. "He still has 50 mg of caffeine" is.

## Remember from the lesson

**Half-life:** the time it takes for half of something to disappear.

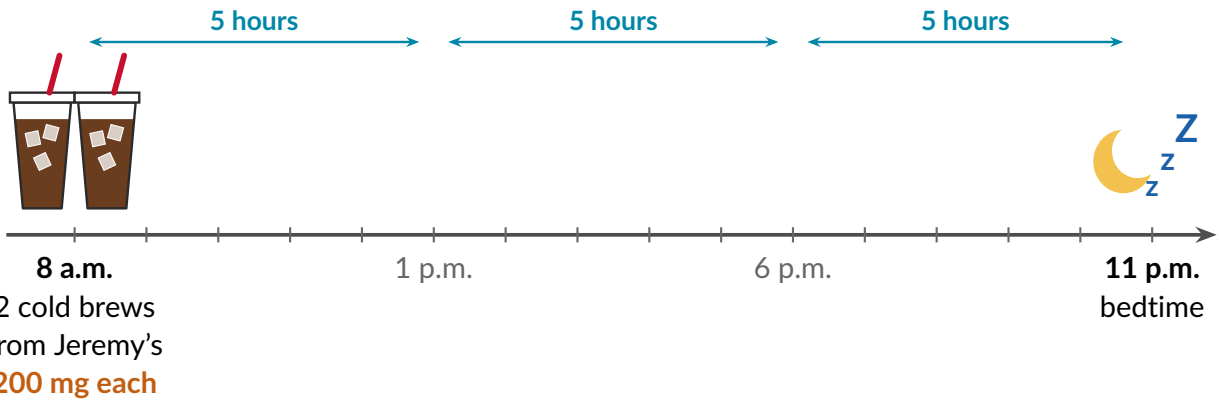
**Carbon-14:** a rare kind of carbon that slowly breaks down into nitrogen.

**Carbon dating:** finding the age of something that was once alive by measuring how much carbon-14 is left.

Caffeine's half-life: about **5 hours** Carbon-14's half-life: **5 700 years**

## Part A Mr Bowen's Coffee

On Monday, Mr Bowen drinks two cold brews at 8 a.m. He goes to bed at 11 p.m.

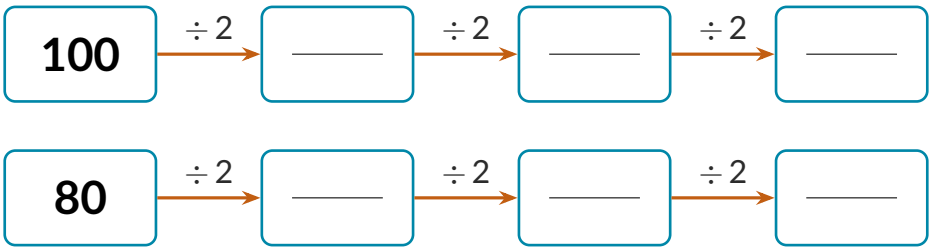


**Word help**

| | |
|---|---|
| **caffeine** | a chemical in coffee, tea, cola and energy drinks. It keeps you awake. |
| **mg** | milligram. A grain of rice is about 20 mg. |
| **cold brew** | strong coffee made slowly with cold water |
| **halve** | cut in half: divide by 2 |
| **curve** | a line that bends smoothly |

## A1 Halving

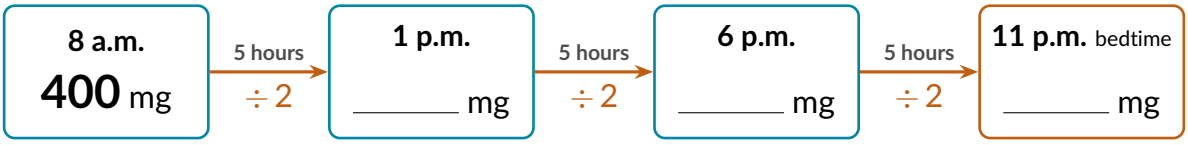
Halve each number.




## A2 Mr Bowen's caffeine

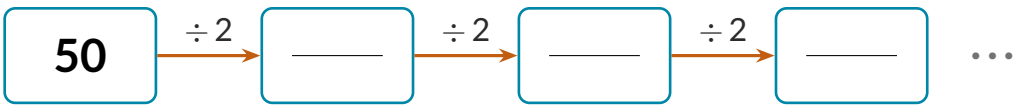
(a) Two cold brews: $2 \times 200 =$ _____ mg

(b) Fill in the boxes. Every 5 hours, **halve** the caffeine.




## A3 Does it ever run out?

(a) Keep halving after bedtime.




(b) If you keep halving, does the number ever get to 0? ☐ yes ☐ no

(c) Fill in the gap with one word from the box.

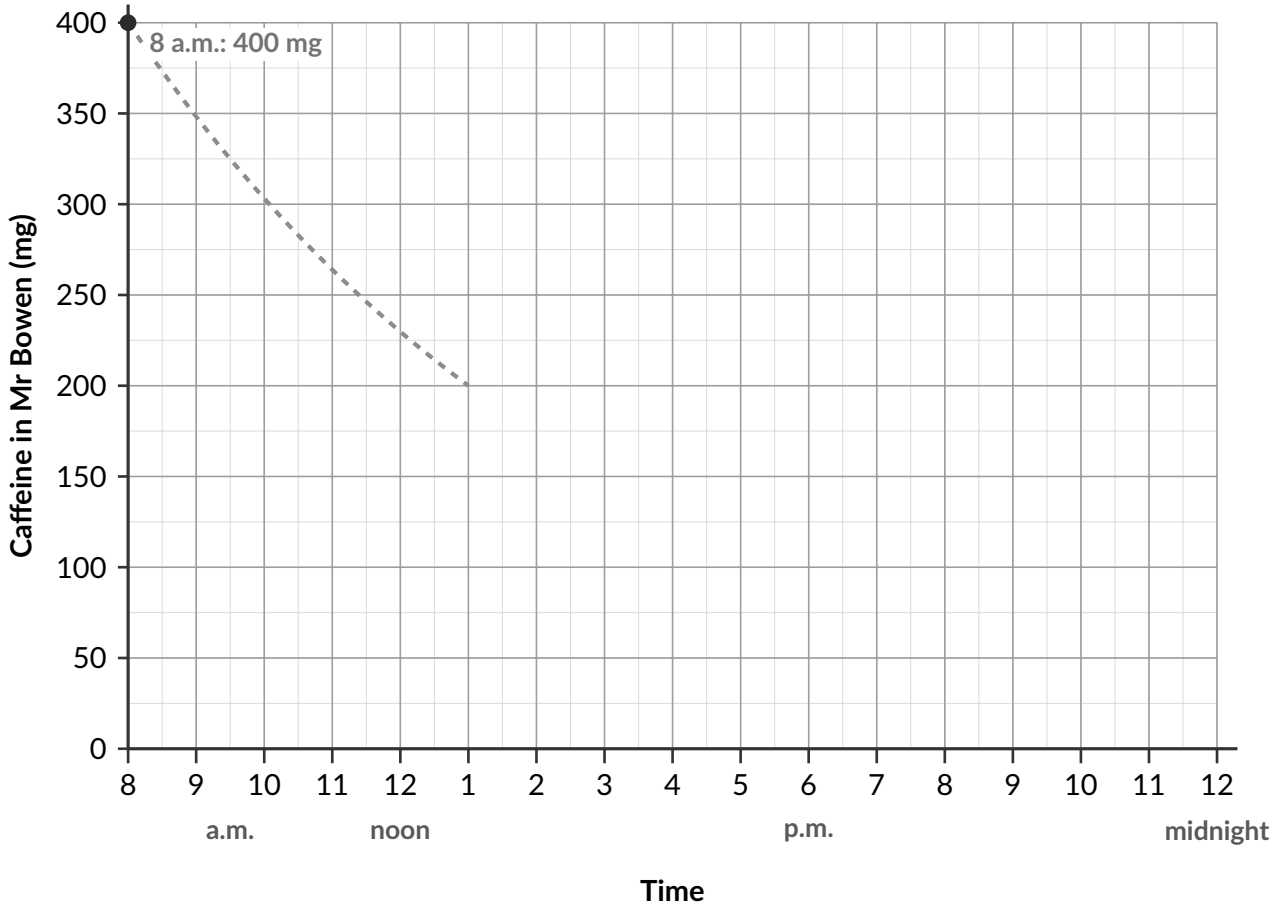
| | | |
|---|---|---|
| half-life | limitation | molecule |

The maths model says the caffeine **never** runs out. But real caffeine is made of molecules, and the last one is gone after about **2 weeks**. So the model is wrong here.

This is a _____ of the model.

## A4 Draw the graph

Put a dot for every box in A2. The first part is drawn for you. Join the dots with a **smooth curve**, not straight lines.




## A5 Read the graph

Mr Bowen marks your homework at **8 p.m.**

(a) 8 p.m. is between 6 p.m. and 11 p.m. So his caffeine is between _____ mg and _____ mg.

(b) Go **up** from 8 p.m. to your curve, then **across**. How much caffeine is in him at 8 p.m.? about _____ mg

## A6 The title question

Why is Mr Bowen still awake at 11 p.m.? Use your number from A2. A cup of coffee has about 100 mg. How much coffee is that?

Mr Bowen is still awake because at 11 p.m. he still has …

# Part B Interview a Caffeine Drinker

**What you will do**

**Interview** an adult in your family who drinks coffee, tea, cola or energy drinks like Sting. Then work out how much caffeine is in their body at **bedtime**.

You can do the interview in **any language**. Write in **English**.

**Word help**

| | |
|---|---|
| **interview** | ask someone questions, and write down the answers |
| **label** | the words printed on a can or bottle. They say what is inside. |
| **search** | type words into Google to find something |
| **estimate** | a careful guess, using numbers |

## B1 The interview

Ask about **yesterday**.

1. "Can I interview you for my homework?"
2. "What was your **first** drink with caffeine yesterday? How big was it?"
   For example: 1 can of Sting, 330 ml.
3. "What time did you drink it?"
4. "What time did you go to bed?"

**I interviewed my** _____.

**Their first drink was** _____

**They drank it at** _____.

**They went to bed at** _____.

## B2 Find the caffeine

**Where to look**

1. **The label** on the can or bottle. Look for the word **caffeine**.
2. **Google.** Search for the drink, like this:
   *caffeine in 1 can of Sting mg*

Milk, sugar and ice: **0 mg**.

Two numbers, like 50 to 80 mg? Use the **middle**: 65 mg.



**Some labels give the caffeine in 100 ml.** This bottle has 20 mg in every 100 ml. It holds 300 ml.
300 ml is 3 lots of 100 ml. $3 \times 20 =$ **60 mg** in the whole bottle.

A 330 ml can? $330 \div 100 = 3.3$, then $3.3 \times 20 = 66$ mg. Or search on Google for the whole can.

Their drink has about _____ mg of caffeine.

I found it on: ☐ the label ☐ Google ☐ another website

## B3 Halve it, every 5 hours

Start with their drink. In each box, **add 5 hours** to the time and **halve** the caffeine. Keep going until you are **past their bedtime**.

**Example: a cup of tea, about 40 mg, at 7 a.m.**

| | The drink | 5 hours later | 5 hours later | 5 hours later | 5 hours later |
|---|---|---|---|---|---|
| **Time** | 7 a.m. | 12 noon | 5 p.m. | 10 p.m. | 3 a.m. |
| **Caffeine (mg)** | 40 | 20 | 10 | 5 | 2.5 |

**Now yours:**

| | The drink | 5 hours later | 5 hours later | 5 hours later | 5 hours later |
|---|---|---|---|---|---|
| **Time** | | | | | |
| **Caffeine (mg)** | | | | | |

## B4 Bedtime

(a) Their bedtime is between the times in two of your boxes: _____ and _____.

(b) So at bedtime, they have between _____ mg and _____ mg.

(c) Choose a number in between. At bedtime, they have about _____ mg of caffeine.

## B5 Compare with Mr Bowen

Mr Bowen had 50 mg at bedtime. Did your adult have **more** or **less**? Why? Think about how **big** the drink was, and **what time** they drank it.

At bedtime my _____ had about _____ mg of caffeine. That is more / less than Mr Bowen, because …

## B6 Is your number right?

Your bedtime number is an **estimate**. Tick **one** reason it might be wrong. Then write it as a sentence.

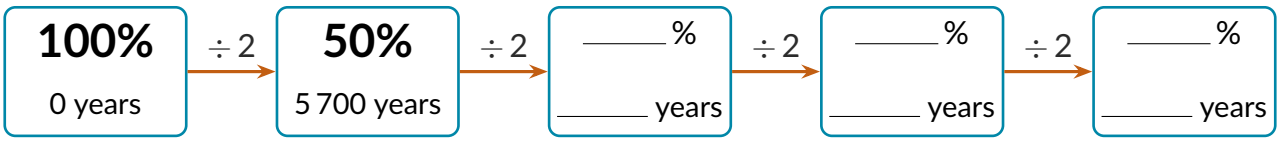
- ☐ They had other drinks with caffeine too.
- ☐ Caffeine's half-life is not 5 hours for everybody.
- ☐ The number from the label or Google may not be exact.

My bedtime number might be wrong because …

# Part C Carbon Dating

## C1 The halving chain

Caffeine halves every 5 hours. Carbon-14 halves every **5 700 years**. Fill in the boxes.




Each box is 5 700 years more than the one before it.

## C2 The museum

Scientists measured the carbon-14 left in five objects. Use C1.

**Careful:** carbon dating only works for things that were **alive**, up to about **50 000 years** old. If it does not work, write **does not work**.

| Object | Was it ever alive? | Carbon-14 left | How many half-lives? | Age (years) |
|---|---|---|---|---|
| A wooden boat | yes | 50% | 1 | 5 700 |
| A bear bone from a cave | | 25% | | |
| A wooden spear | | 12.5% | | |
| A stone axe | | — | | |
| A *T. rex* tooth | | 0% | | |

**Word help:** a **spear** is a long stick with a sharp point · an **axe** is a tool for cutting wood · a ***T. rex*** is a dinosaur

## C3 Why does it not work?

(a) Carbon dating does not work for the stone axe, because stone was never _____ .

(b) Carbon dating does not work for the *T. rex* tooth, because it is too _____ .

*Clue:* dinosaurs died out about 66 million years ago.

# Part D The Sun: Different Speeds, the Same Maths

**Remember from the lesson**

**Caffeine**, **carbon-14** and the **Sun's hydrogen** all disappear at a rate we can model with maths. Measure what is **left** to find **how long** it has been.

**Word help:** the **core** is the middle of the Sun · **hydrogen** and **helium** are gases · a **billion** is a thousand million

## D1 How old is the Sun?

Nobody can take a piece of the Sun. So scientists build a **model Sun** with maths, and let it get older until it matches the real Sun.

(a) The Sun turns _____ into _____ .

(b) When the Sun was born, its core was **71%** hydrogen. Half of 71% is _____ %.

(c) Today the core is **34%** hydrogen. Is about **half** of the hydrogen gone? ☐ yes ☐ no

(d) The model Sun matches the real Sun at about _____ billion years. So the Sun is about _____ billion years old.

## D2 Three clocks

Draw a line from each clock to how fast it goes.

| | | |
|---|---|---|
|  **Caffeine** | • | • half gone every 5 700 years |
|  **Carbon-14** | • | • about half gone in 4.6 billion years |
|  **Hydrogen in the Sun's core** | • | • half gone every 5 hours |

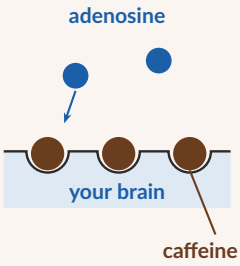
Put them in order, **fastest** first.

1. _____ 2. _____ 3. _____

# Part E Caffeine and Sleep

**Read this: how caffeine affects sleep**

1. All day your brain makes **adenosine** (a-DEN-o-seen), a chemical that makes you feel sleepy.
2. **Caffeine blocks adenosine.** The adenosine is still there, but your brain cannot feel it.
3. Caffeine makes it harder to **fall asleep** and **stay asleep**.
4. **How much is too much?** Even **100 mg** (one cup of coffee) near bedtime can stop an adult sleeping. A child: about **50 mg**.
5. **Stop caffeine 8 hours before bed**, or longer after a big drink.
6. At your age you need **9 to 11 hours** of sleep.



Caffeine sits in the gaps. Adenosine cannot get in.

## E1 Plan your letter

(a) At 11 p.m. Mr Bowen still has _____ mg of caffeine. Caffeine blocks _____.

(b) Bed is at 11 p.m. One cup of coffee must be before _____.

(c) **One** cold brew (200 mg) at 8 a.m.: 200 → _____ → _____ → _____ mg at 11 p.m.

## E2 A letter to Mr Bowen

Write to Mr Bowen: **why** is he still awake, and **how** can he sleep better? Tick each one:

- ☐ Start with *Dear Mr Bowen,*
- ☐ The word **half-life**
- ☐ Two numbers from E1
- ☐ One fact from the guide
- ☐ One piece of advice, and **why**
- ☐ At least **5 full sentences**

## Answer Key — Teacher Copy

The only maths on this sheet is halving and ×5 700: the deck's own. Calculators are allowed. Mr Bowen's times are exactly 5 hours apart, so Part A has no rounding.

### A1 Halving

100: **50, 25, 12.5** · 80: **40, 20, 10**

Expect 12 or 13 for 12.5 from a student avoiding the decimal. Push for 12.5: it is the deck's 10 p.m. number.

### A2 Mr Bowen's caffeine

(a) **400** mg · (b) 1 p.m. **200** · 6 p.m. **100** · 11 p.m. **50**

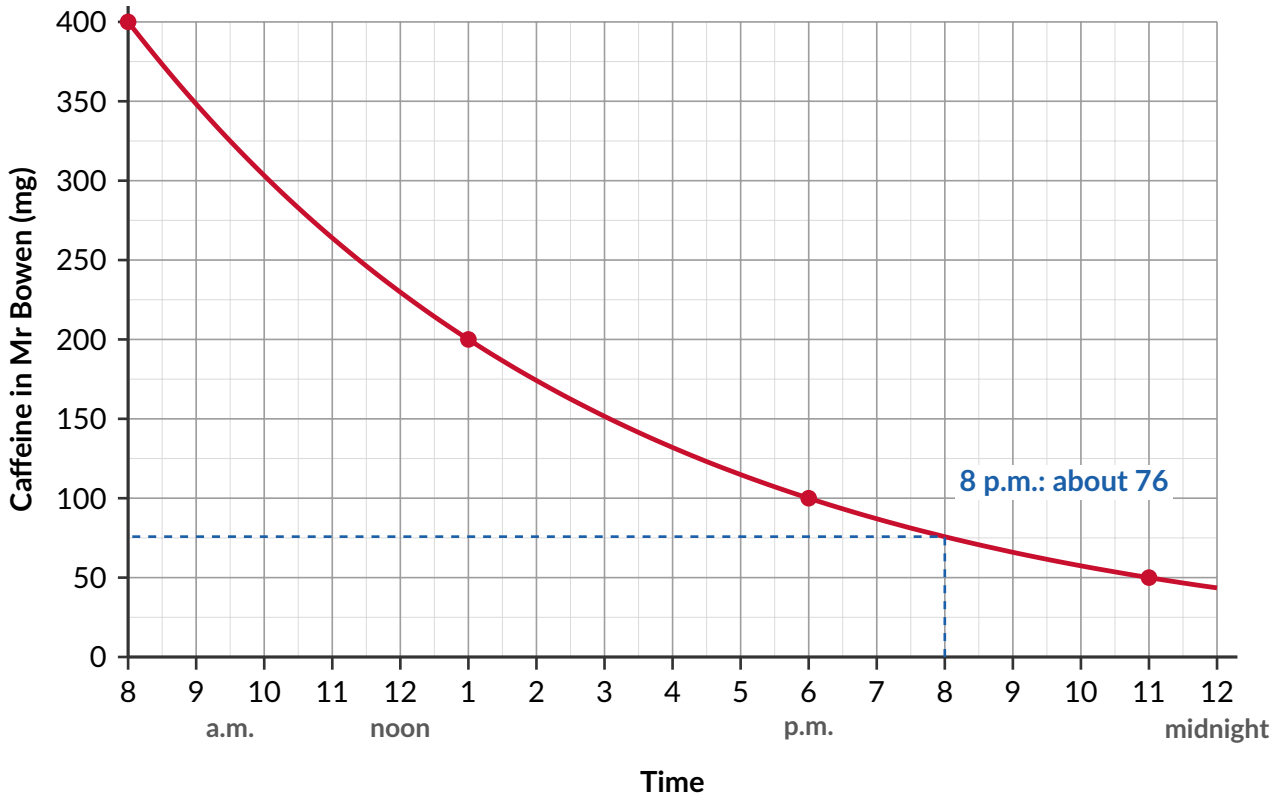
**The error to catch: 200, 0, 0.** That student took away 200 every time — half of the *first* number — instead of halving what is *left*. Ask: half of what? The caffeine is then all gone by 6 p.m., and A3 says the halving model never runs out.

### A3 Does it ever run out?

(a) **25, 12.5, 6.25** · (b) **no** · (c) **limitation**

Deck slide 6. A student who ticks "yes" has thought about real caffeine, not the model: ask what the *maths* says, then read (c) together.

### A4 The graph




Look for a **curve**. Straight lines between the dots are the common one; they make A5(b) come out at 80 instead of 76.

## A5 Read the graph

(a) between **100** and **50** mg · (b) about **76** mg (accept 70–85)

(a) is deck slide 5's method, said in words: name the two points either side before reading between them. 80 is fine from a student who drew straight lines; ask them why the curve sits a little lower.

## A6 The title question

*Mr Bowen is still awake because at 11 p.m. he still has 50 mg of caffeine in his body. That is half a cup of coffee.*

Check for the number *and* the comparison. "Because he drank coffee" is true and is not the answer.

## B1–B4 The interview and the halving table

No single answer. Mark in this order, and stop at the first thing that is missing — it is the thing to send back:

1. A real person, named by relationship ("my mother"), one drink with its size and time, and a bedtime.
2. B2: an amount in mg and a ticked source. A blank is not acceptable; any honest number is.
3. B3 starts at the drink's time, adds 5 hours a box, and halves each time. **Check the clock**: 9 a.m. + 5 hours is 2 p.m., not "14" or "14 p.m."; 8 p.m. + 5 hours is 1 a.m. The tea example crosses noon and midnight on purpose.
4. B4: the two boxes either side of bedtime, their two numbers, and a number between them. If bedtime falls exactly on a box, that box is the answer.

**Numbers to sanity-check against** (all approximate; the point is that the student found one and said where): Sting, 330 ml, about 50–85 mg (the Vietnamese label gives a range per litre, worth a minute in class) · cola, 330 ml can, about 30–35 mg · Red Bull, 250 ml can, about 50–80 mg depending on the version · one espresso shot, about 60–75 mg · Vietnamese iced coffee, about 100–200 mg (robusta, and very variable) · a cup of black or green tea, about 30–50 mg.

## B5 Compare with Mr Bowen

Needs a number, more or less, and *because* with a reason from their own table: a bigger drink, a smaller drink, or a drink earlier or later in the day. Most adults who drink one morning coffee will be well under 50 mg; an afternoon Sting will not. That sheet is worth showing the class (without the name).

## B6 Is your number right?

Any of the three, written as a sentence. "I might have made a mistake" is not a limitation of the model — ask which part of the model. A student who adds a reason of their own (the caffeine takes time to get into the blood; yesterday was not a normal day) deserves the credit.

## C1 The halving chain

100%, 50%, **25%, 12.5%, 6.25%** · 0, 5 700, **11 400, 17 100, 22 800** years

## C2 The museum

| | | | | |
|---|---|---|---|---|
| bear bone | yes | 25% | 2 | **11 400** |
| wooden spear | yes | 12.5% | 3 | **17 100** |
| stone axe | **no** | — | — | **does not work** |
| *T. rex* tooth | **yes** | 0% | — | **does not work** (too old) |

**The *T. rex* is the question.** Expect "no" under *Was it ever alive?* because it is dead now. It *was* alive; it is too old (deck slide 19: 66 million years, over 11 000 half-lives). A student who says "it was alive, so it works" has missed the 50 000-year limit.

## C3 Why does it not work?

(a) **alive** · (b) **old**

## D1 How old is the Sun?

(a) **hydrogen** into **helium** · (b) **35.5%** · (c) **yes** · (d) about **4.6**; about **4.6**

(c): 34 is close to 35.5, so about half is gone — deck slide 21's hydrogen bar. Do not let "half-life of 4.6 billion years" creep in: the Sun burns its hydrogen as fuel; it does not decay. The deck says "about half gone in 4.6 billion years", and so does this sheet. (b) is there because "half of 71" is a real step; 35 or 36 is fine.

## D2 Three clocks

Caffeine → **every 5 hours** · Carbon-14 → **every 5 700 years** · the Sun → **4.6 billion years** · order: **caffeine, carbon-14, the Sun**

## E1 Plan your letter

(a) **50** mg; **adenosine** · (b) **3 p.m.** · (c) **100, 50, 25** mg

(b) is 11 p.m. minus 8 hours; expect 4 p.m. or 2 p.m. from a student counting on their fingers. (c) is the advice with a number in it: one cold brew instead of two halves his bedtime caffeine, 50 mg to 25 mg. A student who writes 100 at 11 p.m. halved once for the whole day.

**Expect: "50 mg is less than 100 mg, so he should sleep."** Good thinking, and worth a minute in class. The 100 mg is a cup drunk *near* bedtime, and it is "can", not "will": some adults are kept awake by less. His real problem is the size of the drink: 400 mg is four cups, so 15 hours was not enough — the guide's "a bigger drink needs longer".

## E2 A letter to Mr Bowen

No sentence starter on purpose: the students begin it themselves. Mark the six ticks in order; the first one missing is the thing to send back. The advice must come with a *why*. A model answer:

*Dear Mr Bowen, You drank 400 mg of caffeine at 8 a.m. That is four cups of coffee. Caffeine has a half-life of 5 hours, so at 11 p.m. you still have 50 mg in your body. Caffeine blocks adenosine, so your brain does not feel sleepy. Please drink only one cold brew, because then you will have 25 mg at bedtime, not 50 mg. From, …*

**The guide's facts, and where they come from.** (1–2) Adenosine builds up while we are awake and makes us sleepy; caffeine blocks its receptors. (3) Caffeine delays sleep and adds waking in the night. (4) European Food Safety Authority, *Scientific Opinion on the safety of caffeine* (2015): single doses of 100 mg (1.4 mg per kg for a 70 kg adult) may delay sleep and shorten it in some adults, especially close to bedtime. The 50 mg for a child is our arithmetic, not an EFSA figure: the same 1.4 mg per kg for a 35–40 kg Year 7. For the whole day, Health Canada's maximum for ages 10–12 is 85 mg. (5) Gardiner et al., *Sleep Medicine Reviews* 2023, a meta-analysis: a cup of coffee (about 107 mg) at least 8.8 hours before bed to avoid losing sleep, and a 217 mg dose at least 13.2 hours. Drake et al. (2013) found 400 mg even 6 hours before bed cut sleep by more than an hour, which is why the sheet says 8, not the 6 often quoted. (6) US National Sleep Foundation: 9–11 hours for ages 6–13 (the American Academy of Sleep Medicine says 9–12 to age 12, 8–10 from 13).